

A Log-Linear Regression Analysis of Allometric Scaling and Species Effects on Penguin Body Mass

Avigyan Chakraborty

Roll No.:

April, 2026

Abstract

The latent-grade model uses the trimmed file `nodules_train_trimmed.csv` rather than the entire training data: we removed individual ratings at least two grades from their nodule's median (212 of the 4647 training ratings, 4.6%, across 204 nodules, one of which lost all of its ratings). The naive models and the test set use all ratings. This rule was chosen after an initial fit gave implausible results. It improves the interpretability of the latent classes, but it reduces measured disagreement, so the estimated confusion matrix overstates reader reliability and predictive fit on the untrimmed ratings is worse.

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1 Introduction

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2 Data Description

The dataset used in this study is the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) database[1]. It contains thoracic CT scans in which up to four radiologists outlined each lung nodule of at least 3 mm and rated its appearance and its likely malignancy.

The unit of analysis is the *nodule*. Geometric features are averaged across a nodule's readers and semantic ratings are summarised by their median. We use the median because the ratings are ordinal, so the gaps between grades have no fixed numerical meaning, and because it is less sensitive to a single outlying reader than the mean.

The variables used in this study are as follows.

Response

- **Malignancy (`mal_median`)**: The median of the readers' ratings, on an ordered scale from 1 (highly unlikely to be malignant) to 5 (highly suspicious). A median falling between two grades is rounded up, which affects 548 nodules (20.8%). Each reader's rating is also retained for the latent-grade model. Discrete (ordinal), with support $\{1, 2, 3, 4, 5\}$.

Primary model: geometric features These are computed from the outlines drawn by the radiologists and are treated as continuous random variables.

- **Log Volume (`log_volume`)**: The natural logarithm of the nodule's volume in mm^3. The logarithm is used because volume is strongly right-skewed. Continuous, with support $\mathbb{R}$.
- **Elongation (`elongation`)**: A value near 1 means that the nodule is not elongated. Continuous, with support $[0, 1]$.
- **Flatness (`flatness`)**: A value near 1 means that the nodule is not flat. Continuous, with support $[0, 1]$ and a point mass at 0.
- **Solidity (`solidity`)**: The number of voxels in the nodule divided by the number in its convex hull. A low value indicates an irregular outline. Continuous, with

support $(0, 1]$.

- **Compactness (`compactness`)**: $V^{2/3}/S$, for volume V and surface area S. A high value means little surface area relative to volume. Continuous, with support $(0, \infty)$.

Secondary model: primary features and semantic ratings These are ordinal ratings from 1 to 5, given by the same radiologists who rate malignancy. The secondary model adds them to the five geometric features above. Since each is a median across readers, all five are discrete (ordinal), with support $\{1, 1.5, 2, \ldots, 4.5, 5\}$.

- **Spiculation (`spiculation`)**: Spiky extensions from the edge, from none to marked.
- **Lobulation (`lobulation`)**: Rounded bulges along the edge, from none to marked.
- **Margin (`margin`)**: Definition of the edge, from poorly defined to sharp.
- **Texture (`texture`)**: Internal density, from non-solid to solid.
- **Sphericity (`sphericity`)**: Overall roundness, from linear to round.

The data are split by patient into a training set (609 patients, 1809 nodules) and a test set (261 patients, 828 nodules). Flatness is exactly 0 for 45 training nodules whose outlines lie in a single plane.

2.1 Summary Statistics

All summary statistics are computed on the training set. Table 1 gives the unstandardised values of the predictors, and Table 2 breaks them down by median malignancy grade.

Model	Variable	Mean	Minimum	Maximum	Std. Dev.
Primary	Log Volume	5.118	2.442	10.345	1.394
	Elongation	0.753	0.306	0.988	0.121
	Flatness	0.555	0.000	0.921	0.164
	Solidity	0.795	0.228	1.000	0.106
	Compactness	0.177	0.058	0.312	0.044
Secondary	Spiculation	1.476	1.000	5.000	0.889
	Lobulation	1.572	1.000	5.000	0.883
	Margin	3.899	1.000	5.000	1.138
	Texture	4.363	1.000	5.000	1.202
	Sphericity	3.789	1.000	5.000	0.826

Table 1: Overall Summary Statistics ($n = 1809$ training nodules)

Model	Variable	Grade 1 (233)	Grade 2 (378)	Grade 3 (823)	Grade 4 (239)	Grade 5 (136)
Primary	**Log Volume**	4.563 (0.981)	4.474 (0.935)	4.779 (0.983)	6.349 (1.269)	7.744 (1.082)
	Elongation	0.749 (0.135)	0.731 (0.125)	0.758 (0.117)	0.773 (0.115)	0.761 (0.109)
	Flatness	0.591 (0.150)	0.511 (0.173)	0.548 (0.167)	0.584 (0.151)	0.602 (0.132)
	Solidity	0.842 (0.092)	0.812 (0.092)	0.808 (0.093)	0.742 (0.104)	0.676 (0.125)
	Compactness	0.184 (0.049)	0.187 (0.043)	0.183 (0.040)	0.154 (0.037)	0.135 (0.031)
Secondary	**Spiculation**	1.172 (0.645)	1.200 (0.632)	1.394 (0.766)	1.795 (0.930)	2.699 (1.243)
	Lobulation	1.255 (0.700)	1.300 (0.728)	1.509 (0.819)	1.933 (0.878)	2.618 (0.970)
	Margin	4.674 (0.793)	3.971 (1.148)	3.756 (1.159)	3.669 (1.076)	3.643 (1.011)
	Texture	4.880 (0.560)	4.344 (1.253)	4.197 (1.323)	4.324 (1.139)	4.603 (0.893)
	Sphericity	4.043 (0.924)	3.810 (0.877)	3.779 (0.781)	3.653 (0.790)	3.599 (0.735)

Table 2: Mean (Std. Dev.) by Median Malignancy Grade (number of training nodules in brackets)

2.2 Data Source

The dataset used in this study is the LIDC-IDRI database[1], hosted by The Cancer Imaging Archive[2] at LIDC-IDRI (TCIA). The full image collection is about 133 GB, but the radiologists' annotations are bundled, already parsed, with the Python library `pylidc`[4]. Every quantity used in this project is computed from these annotations with `pylidc` version 0.2.3, so no CT images were downloaded.

3 Model Specification

3.1 Logistic Regression Model (Naive Proportional Odds)

Methodology Let $Y_i \in \{1, \dots, 5\}$ be the median malignancy grade of nodule i, and let x_i be its vector of predictors. Since Y_i is ordered, we model its cumulative probabilities with a proportional odds model (an ordinal logistic regression):

$$\operatorname{logit} P(Y_i \le k \mid x_i) = \alpha_k - x_i^{\mathsf{T}} \beta, \qquad k = 1, \dots, 4,$$

where $\alpha_1 < \cdots < \alpha_4$ are thresholds. A positive β_j means that a larger x_{ij} shifts a nodule towards higher grades. The odds of being above any given grade are multiplied by e^{β_j} for each one-unit increase in x_{ij}, and this factor is assumed to be the same at every threshold. The model is *naive* because it takes the median as the true grade and ignores disagreement between readers.

Writing $F(t) = e^t/(1 + e^t)$ for the logistic distribution function, with $\alpha_0 = -\infty$ and

$\alpha_5 = \infty$, and treating nodules as independent, the likelihood is

$$L(\alpha, \beta) = \prod_{i=1}^{n} \prod_{k=1}^{5} \left[F(\alpha_k - x_i^\top \beta) - F(\alpha_{k-1} - x_i^\top \beta) \right]^{\mathbf{1}\{Y_i = k\}}.$$

The models are fitted by maximising L on the training set with `polr`. All predictors are standardised, so one unit is one standard deviation. Confidence intervals for the odds ratios are profile-likelihood intervals, obtained by inverting the likelihood ratio test of $H_0 \colon \beta_j = b$ using the χ^2_1 limit given by Wilks' theorem, and then exponentiating the endpoints.

Wilks' theorem states that, under regularity conditions and if H_0 is true, the likelihood ratio statistic satisfies

$$\Lambda = 2 \Big[\sup_{\theta \in \Theta} \ell(\theta) - \sup_{\theta \in \Theta_0} \ell(\theta) \Big] \xrightarrow{d} \chi^2_r$$

as $n \to \infty$, where ℓ is the log-likelihood and r is the number of parameters fixed by H_0. This is a large-sample result. Here $r = 1$, so the approximate 95% interval for β_j is the set of values b with $\Lambda(b) \le \chi^2_{1,0.95} = 3.841$.

Primary Model The fitted model is

$$\begin{aligned} \operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(& 1.330 \cdot \texttt{log_volume} + 0.333 \cdot \texttt{elongation} \\ & - 0.173 \cdot \texttt{flatness} - 0.342 \cdot \texttt{solidity} \\ & + 0.047 \cdot \texttt{compactness} \big), \end{aligned}$$

with $\hat{\alpha} = (-2.503, -1.108, 1.674, 3.488)$ and maximised log-likelihood -2155.134. The estimates are given in Table 3.

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
`log_volume`	1.330	0.065	3.781	(3.335, 4.296)
`elongation`	0.333	0.055	1.395	(1.253, 1.555)
`flatness`	-0.173	0.053	0.841	(0.757, 0.934)
`solidity`	-0.342	0.065	0.710	(0.625, 0.806)
`compactness`	0.047	0.059	1.048	(0.934, 1.175)

Table 3: Primary Proportional Odds Model

Interpretation Each odds ratio is for a one standard deviation increase in that predictor, holding the others constant.

- **Size:** Log volume has by far the largest effect. One standard deviation, roughly

a fourfold increase in volume, multiplies the odds of a higher malignancy grade by 3.78.

- **Solidity:** A more solid nodule has lower odds of a higher grade (odds ratio 0.71), so irregular outlines are rated as more suspicious.
- **Elongation and Flatness:** Elongation is close to 1 for a nodule that is not elongated, so an odds ratio of 1.40 means that more elongated nodules have lower odds of a higher grade. Flatness works in the other direction: its odds ratio of 0.84 means that flatter nodules have higher odds.
- **Compactness:** The confidence interval contains 1, so the data are compatible with no effect once the other predictors are included.

Secondary Model The geometric features only describe the outline of a nodule, whereas radiologists also judge its appearance on the scan. We therefore add the semantic ratings to measure how much this assessment explains beyond geometry: spiculation, lobulation and margin describe the edge in terms that radiologists themselves use, texture records internal density, which the outlines cannot capture, and sphericity gives a reader's view of overall shape. Adding these ratings gives

$$\begin{aligned}\operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(&1.210 \cdot \texttt{log_volume} + 0.321 \cdot \texttt{elongation} \\ &- 0.065 \cdot \texttt{flatness} - 0.160 \cdot \texttt{solidity} \\ &+ 0.002 \cdot \texttt{compactness} + 0.346 \cdot \texttt{spiculation} \\ &+ 0.091 \cdot \texttt{lobulation} - 0.280 \cdot \texttt{margin} \\ &+ 0.063 \cdot \texttt{texture} + 0.002 \cdot \texttt{sphericity}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.572, -1.144, 1.735, 3.617)$ and maximised log-likelihood -2108.642. The estimates are given in Table 4.

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
log_volume	1.210	0.067	3.352	(2.942, 3.828)
elongation	0.321	0.056	1.379	(1.237, 1.539)
flatness	-0.065	0.056	0.937	(0.840, 1.045)
solidity	-0.160	0.069	0.852	(0.744, 0.976)
compactness	0.002	0.059	1.002	(0.893, 1.126)
spiculation	0.346	0.067	1.413	(1.241, 1.613)
lobulation	0.091	0.066	1.096	(0.964, 1.246)
margin	-0.280	0.074	0.756	(0.654, 0.874)
texture	0.063	0.067	1.065	(0.933, 1.215)
sphericity	0.002	0.051	1.002	(0.907, 1.107)

Table 4: Secondary Proportional Odds Model

Interpretation

- **Semantic Ratings:** More spiculated nodules have higher odds of a higher grade (odds ratio 1.41), and nodules with sharper margins have lower odds (0.76). The confidence intervals for lobulation, texture and sphericity contain 1.
- **Geometric Features:** Size remains the largest effect (odds ratio 3.35), and elongation barely changes (1.38). The solidity effect moves towards 1 (0.85), and the interval for flatness now contains 1, which suggests that the ratings capture some of the same information about shape.
- **Circularity:** The semantic ratings come from the same radiologists who rate malignancy, so these associations may partly reflect a reader's overall impression of a nodule.

3.2 Latent True-Grade Model (EM Algorithm)

Methodology The naive model treats the median rating as the true grade. Instead, we now treat the true grade $Z_i \in \{1, \ldots, 5\}$ of nodule i as an unobserved variable, and the readers' ratings R_{ij} as noisy measurements of it. The model has two parts:

- **Structural Model:** The proportional odds model now describes the latent grade, using the primary features x_i:

$$p_{ik}(\alpha, \beta) = P(Z_i = k \mid x_i) = F(\alpha_k - x_i^\mathsf{T}\beta) - F(\alpha_{k-1} - x_i^\mathsf{T}\beta).$$

- **Measurement Model:** A reader rates a nodule of true grade k as r with probabil-

ity $P(R_{ij} = r \mid Z_i = k) = \pi_{kr}$, where $\sum_r \pi_{kr} = 1$. All readers share this 5×5 matrix, because reader identities are not consistent across scans. Ratings are assumed to be conditionally independent given Z_i.

Let $\Psi = (\alpha, \beta, \pi)$, and let $C_{ir} = \#\{j : R_{ij} = r\}$ be the number of readers who gave nodule i the rating r. The ratings are the observed (*incomplete*) data, and their log-likelihood is

$$\ell(\Psi) = \sum_{i=1}^{n} \log \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

The sum inside the logarithm makes this hard to maximise directly. If Z_i were observed, the *complete* log-likelihood would separate into simpler terms:

$$\ell_c(\Psi) = \sum_{i=1}^{n} \sum_{k=1}^{5} \mathbf{1}\{Z_i = k\} \Big[\log p_{ik}(\alpha, \beta) + \sum_{r=1}^{5} C_{ir} \log \pi_{kr} \Big].$$

We therefore estimate Ψ with the EM algorithm[3].

Core Steps Starting from an initial guess $\Psi^{(0)}$, each iteration $t = 0, 1, 2, \ldots$ does the following.

1. **E-step:** Compute the posterior probability of each true grade given the ratings and the current parameters:

$$w_{ik}^{(t)} = P\big(Z_i = k \mid R_i, x_i, \Psi^{(t)}\big) = \frac{p_{ik}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{kr}^{(t)}\big)^{C_{ir}}}{\sum_{l=1}^{5} p_{il}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{lr}^{(t)}\big)^{C_{ir}}}.$$

   Taking the expectation of ℓ_c then replaces $\mathbf{1}\{Z_i = k\}$ with $w_{ik}^{(t)}$, which gives the Q-function:

$$Q\big(\Psi \mid \Psi^{(t)}\big) = \sum_{i=1}^{n} \sum_{k=1}^{5} w_{ik}^{(t)} \log p_{ik}(\alpha, \beta) + \sum_{i=1}^{n} \sum_{k=1}^{5} \sum_{r=1}^{5} w_{ik}^{(t)} C_{ir} \log \pi_{kr}.$$

2. **M-step:** Set $\Psi^{(t+1)} = \operatorname{argmax}_{\Psi} Q(\Psi \mid \Psi^{(t)})$. The two terms of Q involve different parameters, so they are maximised separately:

   - The update for π has a closed form, found using a Lagrange multiplier for the constraint $\sum_r \pi_{kr} = 1$:

$$\pi_{kr}^{(t+1)} = \frac{\sum_i w_{ik}^{(t)} C_{ir}}{\sum_i \sum_s w_{ik}^{(t)} C_{is}}.$$

   - The update for (α, β) has no closed form. It is a proportional odds fit in which each nodule appears once for every grade k, with weight $w_{ik}^{(t)}$, and it is

computed with `polr`.

3. **Stopping Criterion:** Stop when $\left|\ell(\Psi^{(t+1)}) - \ell(\Psi^{(t)})\right| < \epsilon$ for a small threshold ϵ, and take $\hat{\Psi} = \Psi^{(t+1)}$.

EM never decreases the observed log-likelihood, that is, $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$, but it may converge to a local maximum rather than the global one. The result can therefore depend on $\Psi^{(0)}$.

Estimates The algorithm is run on the training set from six starting values, the weights implied by the median rating and five random perturbations of them. All six reach the same log-likelihood, -5190.172, so we find no sign of a local maximum. The fitted structural model is

$$\begin{aligned}\operatorname{logit} \hat{P}(Z_i \leq k \mid x_i) = \hat{\alpha}_k - \big(&1.682 \cdot \texttt{log_volume} + 0.471 \cdot \texttt{elongation} \\ &- 0.323 \cdot \texttt{flatness} - 0.583 \cdot \texttt{solidity} \\ &+ 0.010 \cdot \texttt{compactness}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.983, -0.494, 1.968, 4.024)$. Table 5 compares the odds ratios with the naive primary model, and Table 6 gives the estimated matrix $\hat{\pi}$.

Predictor	$\hat{\beta}$ (EM)	Odds Ratio (EM)	Odds Ratio (Naive)
`log_volume`	1.682	5.375	3.781
`elongation`	0.471	1.601	1.395
`flatness`	-0.323	0.724	0.841
`solidity`	-0.583	0.558	0.710
`compactness`	0.010	1.010	1.048

Table 5: EM Estimates of the Structural Model

Latent Grade	Rated 1	Rated 2	Rated 3	Rated 4	Rated 5
1	0.956	0.044	0.000	0.000	0.000
2	0.103	0.395	0.499	0.002	0.000
3	0.035	0.255	0.541	0.169	0.000
4	0.000	0.092	0.313	0.397	0.197
5	0.004	0.000	0.099	0.194	0.703

Table 6: Estimated Rating Probabilities $\hat{\pi}_{kr}$

Interpretation

- **Structural Model:** The directions of the four main effects match the naive model. Each of these odds ratios is further from 1 under EM, and size remains the largest effect. Compactness is again close to 1. EM does not provide standard errors directly, so these differences are not yet tested.
- **Latent Grades:** Latent grades 1, 3, 4 and 5 are most often rated as themselves, whereas latent grade 2 is rated 3 more often than 2. The latent grades should therefore be read as ordered groups rather than as exact matches to the rating scale.

4 Statistical Inference

4.1 Confidence Intervals

In multiple linear regression, estimating the point values for the coefficients ($\hat{\beta}_i$) provides a single best guess for the population parameters. However, to account for sampling variability and quantify the precision of these estimates, we construct 95% confidence intervals for each predictor, the mean response, and future individual observations.

4.1.1 Coefficient Intervals

The true parameter ($\hat{\beta}_i$) is estimated to fall within the following interval with a 95% level of confidence:

$$\hat{\beta}_i \pm t_{\alpha/2}(n-p-1) \cdot SE(\hat{\beta}_i).$$

Plugging in the values, we derive the following interval estimates:

Predictor Variable	**Estimate**	**Lower Bound** (2.5%)	**Upper Bound** (97.5%)
`Intercept` ($\hat{\beta}_0$)	0.0759	-1.2559	1.4077
log(`Flipper_Length`) ($\hat{\beta}_1$)	0.9036	0.6099	1.1972
log(`Culmen_Length`) ($\hat{\beta}_2$)	0.4364	0.2864	0.5865
log(`Culmen_Depth`) ($\hat{\beta}_3$)	0.6181	0.4598	0.7763
`Is_Gen` ($\hat{\beta}_4$)	0.2341	0.1648	0.3034
`Is_Chin` ($\hat{\beta}_5$)	-0.1195	-0.1592	-0.0798

Table 7: 95% Confidence Intervals for the Regression Parameters

Because both the response variable and the continuous predictors were log-transformed, the coefficients are interpreted as elasticities, representing the expected percentage change in body mass for a 1% increase in the predictor, holding all other variables constant.

- **Flipper Length:** We are 95% confident that for every 1% increase in a penguin's flipper length, its expected body mass increases by between 0.61% and 1.20%.
- **Culmen Length:** We are 95% confident that a 1% increase in culmen length corresponds to an expected body mass increase of between 0.29% and 0.59%.
- **Culmen Depth:** We are 95% confident that a 1% increase in culmen depth is associated with a 0.46% to 0.78% increase in expected body mass.
- **Species Effects (`Is_Gen`):** We are 95% confident that, holding all physical measurements constant, a Gentoo penguin will weigh between 17.9% and 35.4% more than the baseline Adelie species.
- **Species Effects (`Is_Chin`):** We are 95% confident that, holding all physical measurements constant, a Chinstrap penguin will weigh between 7.7% and 14.7% less than the Adelie species.
- **Intercept:** The 95% confidence interval for the intercept ranges from -1.2559 to 1.4077. The intercept corresponds to the expected log body mass of a baseline Adelie penguin when all log-transformed predictors are equal to zero. This implies that the original measurements are equal to 1 (since $\log(1) = 0$), which is not biologically meaningful. Thus, the intercept has limited practical interpretation in this context.

4.1.2 Confidence Intervals for the Mean Response

Let x_0 be a column vector containing a specific set of predictor values. To estimate the expected value $\mathbb{E}[\log(\texttt{Body_M})]$ at x_0, we utilize the matrix X to account for the covariance among predictors. The 95% confidence interval for the mean response is given by:

$$\hat{y}_0 \pm t_{\alpha/2}(n-p-1) \cdot \sqrt{\text{MSE} \cdot x_0^T (X^T X)^{-1} x_0}.$$

Here, MSE is the Mean Squared Error, estimating the error variance σ^2.

Application: Expected Mass Across Species To isolate the structural effects of the categorical variables, we construct a theoretical standardized penguin utilizing the mean of the log-transformed morphological measurements. We then predict the expected $\log(\texttt{Body_M})$ across all three species profiles, keeping these baseline log-dimensions mathematically constant.

Species	Point Estimate	Lower Bound	Upper Bound
Adelie (Baseline)	8.2651	8.2337	8.2964
Chinstrap	8.1456	8.1178	8.1733
Gentoo	8.4992	8.4581	8.5402

Table 8: 95% Confidence Intervals for Mean Response

Observing the Chinstrap and Gentoo confidence bounds, we see parallel shifts relative to the Adelie baseline. Because the input traits are identical, we are 95% confident that the average log body mass of a Gentoo population will always remain strictly higher than an identically proportioned Adelie population, whereas a Chinstrap population will remain strictly lower.

4.1.3 Prediction Intervals

While the confidence interval predicts $\mathbb{E}[\log(\texttt{Body_M})]$ for all penguins with a specific set of traits, the prediction interval predicts the exact $\log(\texttt{Body_M})$ of a single, individual new penguin. Because individual observations naturally vary around the population mean, predicting a single point carries additional irreducible uncertainty. The 95% prediction interval of $\log(\texttt{Body_M})$ is mathematically wider and is expressed as:

$$\hat{y}_0 \pm t_{\alpha/2}(n-p-1) \cdot \sqrt{\text{MSE} \cdot (1 + x_0^T (X^T X)^{-1} x_0)}.$$

The addition of 1 inside the parenthetical term accounts for the variance of the residual error (σ^2) associated with drawing a new, random individual from the population.

Application: Predicting Individual Penguins Using the same theoretical baseline log-dimensions, we now calculate the intervals for randomly sampling a single new penguin from each respective species.

Species	Point Estimate	Lower Bound	Upper Bound
Adelie (Baseline)	8.2651	8.1107	8.4195
Chinstrap	8.1456	7.9918	8.2993
Gentoo	8.4992	8.3425	8.6558

Table 9: 95% Prediction Intervals for New Observations

As mathematically expected, these prediction intervals are significantly wider than their corresponding confidence intervals. This increased width explicitly accounts for the in-

herent random variation (σ^2) between individual penguins, even when they belong to the exact same species and share identical physical measurements.

4.2 Hypothesis Testing

To formally evaluate the explanatory power and structural validity of our multiple log-linear regression model, we conduct three distinct tiers of hypothesis testing.

4.2.1 Overall Model Significance

The global F-test evaluates whether the model as a whole explains a statistically significant portion of the variance in expected log body mass. To do this, we compare the full model against a restricted, intercept-only baseline model.

- **Null Hypothesis (H_0):** $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$. None of the predictors have a significant linear relationship with the response.
- **Alternative Hypothesis (H_1):** At least one $\beta_i \neq 0$ (for $i \in \{1, 2, 3, 4, 5\}$).

The test statistic evaluates the reduction in SSE when all predictors are added to the intercept model:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/\text{df}_{\text{restricted}} - \text{df}_{\text{full}}}{\text{SSE}_{\text{full}}/\text{df}_{\text{full}}}.$$

For the global test, the restricted intercept-only model has $\text{df} = n - 1$, and the full model containing 5 predictors has $\text{df} = n - 6$. Evaluating these degrees of freedom, the formula simplifies to:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/5}{\text{SSE}_{\text{full}}/336}.$$

Calculation and Inference Evaluating the global model yields an F-statistic of $F_0 = 340.7279$, which corresponds to a p-value strictly less than 0.001. Because this p-value is significantly smaller than our significance level ($\alpha = 0.05$), we reject the null hypothesis. We conclude with 95% confidence that the regression model as a whole explains a statistically significant proportion of the variance in expected log body mass.

4.2.2 Individual Parameter Significance

While the global F-test evaluates the entire model, marginal t-tests evaluate the independent contribution of each specific predictor, holding all other variables constant.

- **Null Hypothesis (H_0):** $\beta_i = 0$. The i-th predictor does not significantly affect expected log body mass.
- **Alternative Hypothesis (H_1):** $\beta_i \neq 0$.

The test statistic is calculated by dividing the parameter estimate by its standard error:

$$t_0 = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}.$$

Predictor Variable	**Estimate ($\hat{\beta}_i$)**	**t-statistic**	**p-value**
`Intercept`	0.0759	0.112	0.911
log(`Flipper_Length`)	0.9036	6.053	< 0.001
log(`Culmen_Length`)	0.4364	5.722	< 0.001
log(`Culmen_Depth`)	0.6181	7.682	< 0.001
`Is_Gen`	0.2341	6.641	< 0.001
`Is_Chin`	-0.1195	-5.925	< 0.001

Table 10: Marginal t-Tests for Individual Parameters

Inference As shown in the table, with the exception of the intercept, every morphological and categorical predictor yields a p-value strictly less than 0.001. This strongly rejects the null hypotheses at the $\alpha = 0.05$ level, confirming that each variable is a highly significant independent predictor of log body mass.

4.2.3 Joint Significance of Categorical Variables

To rigorously justify the inclusion of the categorical species indicators, we test their joint significance against a restricted model containing only the physical measurements.

- **Null Hypothesis (H_0):** $\beta_4 = \beta_5 = 0$. The species classifications provide no additional explanatory power beyond physical dimensions.
- **Alternative Hypothesis (H_1):** At least one species parameter is non-zero.

We use the same F-statistic as the global test:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/\text{df}_{\text{restricted}} - \text{df}_{\text{full}}}{\text{SSE}_{\text{full}}/\text{df}_{\text{full}}}.$$

For our specific partial test, the restricted model contains 3 continuous predictors (df $= n - 4$), and the full model adds 2 categorical indicators (df $= n - 6$). Evaluating the

degrees of freedom yields:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/2}{\text{SSE}_{\text{full}}/336}.$$

Calculation and Inference Evaluating the joint addition of the species variables yields an F-statistic of $F_0 = 95.7775$ with a corresponding p-value strictly less than 0.001. Because this p-value is significantly smaller than $\alpha = 0.05$, we firmly reject the null hypothesis. This mathematically suggests that the categorical species indicators provide a highly significant improvement to the model's predictive power, reasonably justifying their inclusion over a model based purely on physical dimensions.

4.3 Model Diagnostics

To conclude our analysis, we must verify both the explanatory power of the model and the structural validity of the Ordinary Least Squares (OLS) assumptions in log-space.

4.3.1 Goodness of Fit

The explanatory power of the model is measured by the proportion of total variance in log(`Body_M`) that is captured by our selected predictors.

- **Multiple R^2 (0.8353):** Our model explains 83.53% of the sample variance in the observed log(`Body_M`).
- **Adjusted R^2 (0.8328):** Because adding parameters mathematically inflates the raw R^2. Our high adjusted R^2 suggests that the inclusion of the categorical species indicators provides a genuine increase in predictive accuracy.

4.3.2 Residual Analysis

A regression model is only mathematically valid if its unobserved error terms (ε) are random and independent.

- **Residuals vs. Fitted:** The plotted residuals display a random scatter clustered tightly around the horizontal zero-line with no distinct funnels or curves. This suggests that the variance of the errors is constant and that the log-linear structural relationship is appears appropriate.
- **Normal Q-Q Plot:** The standardized residuals tightly hug the theoretical 45-degree diagonal line, visually indicating that the error terms are normally dis-

tributed.

4.3.3 Normality Assessment

To rigorously prove the visual assumption of normality established by the Q-Q plot, we apply the Pearson Chi-Square Goodness-of-Fit test to the model's standardized residuals.

By standardizing the residuals ($\frac{\varepsilon-\bar{\varepsilon}}{s}$), we map them directly to a standard Normal distribution curve, $N(0,1)$. We slice this theoretical curve into $k = 10$ regions with equal probabilities, meaning the expected probability of a residual falling into any specific region is exactly $E_i = 10\%$.

We then map our actual observed standardized residuals (O_i) into these regions. The test statistic evaluates the difference between the observed counts and the theoretically expected counts:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

Calculation and Inference Evaluating the standardized residuals across 10 equiprobable bins yields a test statistic of $\chi^2 = 9.6374$. With $k - 3 = 7$ degrees of freedom (accounting for the estimation of the mean and variance), this corresponds to a p-value of 0.2101.

Because this p-value is strictly greater than our significance threshold ($\alpha = 0.05$), we **fail to reject** the null hypothesis. This mathematically suggests that our residuals do not significantly deviate from a normal distribution. Consequently, all confidence intervals, prediction intervals, and hypothesis tests calculated in the preceding sections are reasonably supported.

5 Conclusion

This study developed a multiple linear regression model to explain penguin body mass using flipper length, culmen length, culmen depth, and species. Motivated by allometric scaling, a log transformation was applied to obtain a log-linear model that better captures the underlying relationships.

The fitted model demonstrated strong explanatory power, accounting for about 83.5% of the variation in log body mass. All predictors, except the intercept, were highly significant. Flipper length, culmen length, and culmen depth each showed a positive association with body mass, while the species indicators captured meaningful differences among penguin species.

The inferential results supported the usefulness of the model. The overall F-test confirmed statistical significance, and the joint test showed that species provides additional explanatory power beyond physical measurements. Confidence and prediction intervals were consistent with the fitted model.

Diagnostic analysis indicated that the regression assumptions are reasonably satisfied. Residual plots showed no major patterns, the Q-Q plot suggested approximate normality, and the chi-square test did not detect significant deviations. Overall, the model provides both strong predictive performance and a meaningful representation of the relationship between body mass and morphological traits.

A Appendix

A.1 Dataset

The dataset used in this study is the Palmer Penguins dataset[**?**], originally collected by Kristen B. Gorman and made publicly available by Allison Marie Horst, Alison Presmanes Hill, and Kristen B. Gorman.

It can be accessed through the following sources:

- GitHub: github.com/allisonhorst/palmerpenguins
- Kaggle: kaggle.com/datasets/parulpandey/palmer-archipelago-antarctica-penguin-data

A.2 Code

The code used in this project is divided based on functionality:

- R Code (Computation and Model Fitting): R Code Link
- Python Code (Visualization and Plotting): Python Code Link

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